

provider_live_proof_RESULTS_20260707

CONST-039 Per-Provider Live-Proof Harness — RESULTS (2026-07-07)

Status	HONEST MIXED RESULT. The harness itself is proven genuine and correctly discriminating (real HTTP calls, honest SKIP-on-absence, real errors surfaced verbatim, no fabricated PASS anywhere). Of the 10 CONST-039 providers, 2 are PROVEN LIVE this session (Groq, Mistral); 3 never had a credential configured (OpenAI, Anthropic, xAI — honest SKIP); 2 local providers had no reachable server on this host (Ollama, Llama.cpp — honest SKIP); 3 had a credential PRESENT but a real, non-fabricated failure (DeepSeek — insufficient balance; Gemini — invalid key; OpenRouter — empty completion under finish_reason=length).
Harness	helix_code/internal/llm/provider_live_proof_test.go (build tag providerlive) + provider_live_proof_skip_test.go (always-on unit test of the honest-SKIP contract)
Runs summarized	10 timestamped run-ids under docs/qa/provider_live_proof_20260707T* (list in §5)
Branch	feature/helixllm-full-extension
Command	cd helix_code && go test -tags=providerlive -v -count=1 -run TestProviderLiveProof ./internal/llm/

1. Why this exists (gap-closure)

The pre-existing `ensemble_provider_live_probe_test.go` proves the ENSEMBLE orchestration layer — it drives 4 providers as a GROUP and asserts only an aggregate “≥2 successful members” property, with a suite-level `t.Fatalf` on fewer than 2 that is a FAIL (not an honest SKIP) when the operator has 0 or 1 of those 4 keys configured. `provider_live_proof_test.go` is the missing PER-PROVIDER harness: one independent subtest per CONST-039 provider, each with its own honest SKIP/PASS/FAIL gate and its own captured evidence directory — never conflating “no credential” with “broken.”

2. Anti-bluff mechanism (§11.4.2/§11.4.5)

Every hosted-provider probe sends a freshly `crypto/rand`-generated nonce (`LIVEPROOF-<12 hex chars>`) and asserts the model’s response CONTAINS that exact nonce. A cached, mocked, or hardcoded string cannot possibly contain a token that did not exist until the call executed — this is what makes a PASS here unforgeable. Local providers (Ollama, Llama.cpp) are gated on a real HTTP reachability probe against their own base URL/probe path (Llama.cpp deliberately does NOT reuse its own `IsAvailable()`, which is unconditionally true at construction and is a known-unreliable signal — see the harness source comment). No credential value is ever written to the evidence directory (`request.json/response.json` carry only prompt/model/nonce/content — CONST-042 no-secret-leak); a scan of all 10 evidence trees for OpenAI/Google/Groq-shaped key patterns found zero hits.

3. Per-provider verdict table (final/most-informative run:

`provider_live_proof_20260707T215206Z`)

Provider	Verdict	Detail
Groq	PASS — LIVE	Real HTTP call, model llama-3.1-8b-instant, nonce LIVEPROOF-263bf9c2437a echoed verbatim, 338.9ms, 78 total tokens
Mistral	PASS — LIVE	Real HTTP call, model mistral-small-2603, nonce LIVEPROOF-dfd12f6f5638 echoed

Provider	Verdict	Detail
		verbatim, 402.6ms, 63 total tokens
DeepSeek	FAIL (real, non-fabricated)	Key present; API returned HTTP 402 Insufficient Balance — real DeepSeek account has no funds, not a harness defect
Gemini	FAIL (real, non-fabricated)	Key present; API returned HTTP 400 INVALID_ARGUMENT - API key not valid — real invalid/rotated key, confirmed again in the later ...T215310Z re-check run
OpenRouter	FAIL (real, non-fabricated)	Real HTTP round-trip completed (13.8s, 127 total tokens) but finish_reason=length with empty visible content — the selected model spent its token budget before emitting visible text; nonce could not be echoed
Anthropic	SKIP — honest	No recognised env-var alias set to a non-placeholder value this session
OpenAI	SKIP — honest	No recognised env-var alias set to a non-placeholder value this session
xAI	SKIP — honest	No recognised env-var alias set to a non-placeholder value this session
Ollama	SKIP — honest	GET http://localhost:11434/api/tags unreachable — no local server listening on this host during this run
Llama.cpp	SKIP — honest	

Provider	Verdict	Detail
		GET http://localhost:8080/models unreachable — the coder container publishes 8080/tcp internally but does not expose it on the host loopback in this topology

4. Cross-run consistency check (§11.4.98 re-runability)

Runs ...T213702Z, ...T213736Z, ...T214258Z, ...T214440Z, ...T215302Z (and the single-provider spot-checks ...T214305Z/...T214355Z for OpenAI) were executed with NO hosted-provider credentials configured and NO local server reachable; every one of those runs produced the IDENTICAL all-SKIP pattern shown above for the 8 not-yet-configured providers — confirming the harness’s honest-SKIP gate is deterministic and does not flip to a false PASS/FAIL across repeated invocations. Run ...T215206Z is the only run in this set where hosted credentials were present for DeepSeek/Gemini/Groq/Mistral/OpenRouter; run ...T215310Z is a single-provider Gemini spot-check re-confirming the same invalid-key FAIL after the ...T215206Z finding.

5. Runs summarized (evidence directories, this commit)

- provider_live_proof_20260707T213702Z — all-SKIP baseline
- provider_live_proof_20260707T213736Z — all-SKIP baseline (re-run)
- provider_live_proof_20260707T214258Z — all-SKIP baseline (re-run)
- provider_live_proof_20260707T214305Z — OpenAI-only spot-check, SKIP
- provider_live_proof_20260707T214355Z — OpenAI-only spot-check, SKIP
- provider_live_proof_20260707T214440Z — all-SKIP baseline (re-run)
- provider_live_proof_20260707T215206Z — **full run with credentials present** (see §3)
- provider_live_proof_20260707T215302Z — OpenAI-only spot-check, SKIP
- provider_live_proof_20260707T215310Z — Gemini-only re-check, FAIL (invalid key, confirmed)

6. Honest boundary (§11.4.6) — what is proven vs pending

Proven: the harness makes REAL, unforgeable HTTP calls with a per-run nonce, correctly distinguishes SKIP (no credential / no reachable local server) from FAIL (credential present, real API-level rejection) from PASS (credential present, genuine live completion echoing the nonce), and this behaviour is stable across 9 repeated invocations. Groq and Mistral are PROVEN LIVE end-to-end integrations as of this run.

Pending / operator-actionable (not a harness defect): - DeepSeek: the configured key belongs to an account with insufficient balance — needs a funded account or a different key. - Gemini: the configured key is rejected as invalid by Google's API — needs a valid/rotated key. - OpenRouter: the auto-picked catalogue model exhausted its token budget on a 32-token cap without emitting visible content (`finish_reason=length`) — a follow-up should either raise `MaxTokens` for this probe or pin a non-reasoning model via `PROVIDERLIVE_MODEL_OPENROUTER`. - OpenAI, Anthropic, xAI: never had a credential configured this session — genuinely untested, not proven broken. - Ollama, Llama.cpp: no reachable local server on this host during this run — genuinely untested, not proven broken.

No result in this document is fabricated, cached, or asserted beyond what the captured `request.json/response.json/verdict.txt` evidence in each run directory actually shows.